Vietoris endofunctor for closed relations and its de Vries dual

You can watch the recording of the talk at this link: `https://youtu.be/3tazZqybqjQ`

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Kripke semantics connects modal logic with the Vietoris endofunctor.

$$\begin{tikzcd} \mathsf{Stone} \arrow[r, bend left] \arrow[loop below, "\mathbb{V}"] & \mathsf{BA} \arrow[l, bend left, "\mathrm{op}"] \arrow[loop below, "\mathbb{K}"] \end{tikzcd}$$

Algebras for $\mathbb{K}$ = modal algebras.

Coalgebras for $\mathbb{V}$ = descriptive frames.

This is the coalgebraic approach to modal logic.

$$\begin{tikzcd} \mathsf{Stone} \arrow[r, bend left, "\mathrm{op}"] \arrow[loop below, "\mathbb{V}"] & \mathsf{BA} \arrow[l, bend left] \arrow[loop below, "\mathbb{K}"] \end{tikzcd} \qquad \begin{tikzcd} \mathsf{Stone}^{\mathsf{R}} \arrow[r, bend left, "\mathrm{op}"] \arrow[loop below, "\mathbb{V}^{\mathsf{R}}"] & \mathsf{BA}^{\mathsf{S}} \arrow[l, bend left] \arrow[loop below, dashed, "\mathbb{K}^{?\mathsf{S}}"] \end{tikzcd}$$

1. Stone duality has a natural extension to closed relations [Celani, 2018].

2. Vietoris has a natural extension to closed relations [Goy, Petrișan, Aiguier, 2021]! Which arised in studies of the usual Vietoris on functions.

3. We provide a description of the dual of $\mathbb{V}^{\mathsf{R}}$.

Natural notions $\to$ hope for applications! Our application: a resolution of an open problem on de Vries duality [Bezhanishvili, Bezhanishvili, Harding, 2015], concerning the usual Vietoris on compact Hausdorff spaces and continuous functions.

$$\begin{tikzcd} \mathsf{Stone} \arrow[r, bend left] \arrow[r, phantom, "\mathrm{op}"] & \mathsf{BA} \arrow[l, bend left] \end{tikzcd}$$

Extended by Celani (2018) (see also Kurz, Moshier, Jung, 2023):

$$\begin{tikzcd} \mathsf{Stone}^{\mathsf{R}} \arrow[r, bend left] \arrow[r, phantom, "\mathrm{op}"] & \mathsf{BA}^{\mathsf{S}} \arrow[l, bend left] \end{tikzcd}$$

closed relations subordinations

Closed relation $R\colon X \nrightarrow Y$ is a subset $R \subseteq X \times Y$ that is closed; equivalently, such that

- $R[\text{closed}]$ is closed,
- $R^{-1}[\text{closed}]$ is closed.

Why we got into closed relations:

Every compact Hausdorff space X is a continuous image of a Stone space (e.g., its Gleason cover). So it can be presented via

Stone space + closed equivalence relation.

$$\begin{tikzcd}
\text{Stone} & \ni & \mathcal{G}(X) \arrow[r, tail] \arrow[d, two heads] & \mathcal{G}(Y) \arrow[d, two heads] & \in & \text{Stone} \\
\text{KHaus} & \ni & X \arrow[r, "f"] & Y & \in & \text{KHaus}
\end{tikzcd}$$

Dual of a closed relation:

$$\begin{tikzcd} X \arrow[d, tail, "R"] & \mathrm{Clop}(X) \\ Y & \mathrm{Clop}(Y) \arrow[u, tail, "S"'] \end{tikzcd}$$

For $V \in \mathrm{Clop}(Y)$ and $U \in \mathrm{Clop}(X)$:

$$V \, S \, U \iff R^{-1}[V] \subseteq U$$

Example:

$$\begin{tikzcd} X \arrow[d, tail, "="] & \mathrm{Clop}(X) \\ X & \mathrm{Clop}(X) \arrow[u, tail, "\subseteq"'] \end{tikzcd}$$

Subordination $:=$ a relation $S\colon A \looparrowright B$ such that

$$\left(\bigvee_{i=1}^{n} a_i\right) S \left(\bigwedge_{j=1}^{m} b_j\right) \iff \forall i, j \;\; a_i \; S \; b_j.$$

Theorem (Celani, 2018)

$\mathsf{Stone}^{\mathsf{R}}$ *(closed relations) is dual to* BA^{S} *(subordinations).*

Definition (Vietoris hyperspace)

The Vietoris hyperspace $\mathbb{V}(X)$ of a Stone space X is the set of closed subsets of X, equipped with the topology generated by the following subsets of $\mathbb{V}(X)$, for U clopen of X:

$$\Diamond U := \{K \in \mathbb{V}(X) \mid K \cap U \neq \varnothing\},$$
$$\Box U := \{K \in \mathbb{V}(X) \mid K \subseteq U\}.$$

Vietoris functor on Stone:

$$\mathsf{Stone} \xrightarrow{\mathbb{V}} \mathsf{Stone}$$

$$\begin{tikzcd} X \arrow[d, "f"] & \mathbb{V}(X) \arrow[d, "f[-]"] \\ Y & \mathbb{V}(Y) \end{tikzcd}$$

Extension of $\mathbb{V}$ from Stone to $\mathsf{Stone}^{\mathsf{R}}$ [Goy, Petrișan, Aiguier, 2021]:

$$\mathsf{Stone}^{\mathsf{R}} \xrightarrow{\mathbb{V}^{\mathsf{R}}} \mathsf{Stone}^{\mathsf{R}}$$

$$\begin{tikzcd} X \arrow[d, tail, "R"] \\ Y \end{tikzcd} \qquad \begin{tikzcd} \mathbb{V}(X) \arrow[d, tail, "\mathbb{V}^{\mathsf{R}}(R)"] \\ \mathbb{V}(Y) \end{tikzcd}$$

(Egli-Milner:) For $K \in \mathbb{V}(X)$ and $L \in \mathbb{V}(Y)$,

$$K \; \mathbb{V}^{\mathsf{R}}(R) \; L \iff \begin{cases} \forall x \in K \; \exists y \in L : \; x \, R \, y, \\ \forall y \in L \; \exists x \in K : \; x \, R \, y. \end{cases}$$

It restricts to the usual Vietoris functor on continuous functions.

$$\begin{tikzcd}
\mathsf{Stone}^{\mathsf{R}} \arrow[rr, bend left] \arrow[loop below, "\mathbb{V}^{\mathsf{R}}"] & \mathrm{op} & \mathsf{BA}^{\mathsf{S}} \arrow[ll, bend left] \arrow[loop below, dashed, "?"]
\end{tikzcd}$$

What is the dual of $\mathbb{V}^{\mathsf{R}}$?

$$\mathbb{V}^{\mathsf{R}} \colon \mathsf{Stone}^{\mathsf{R}} \to \mathsf{Stone}^{\mathsf{R}}$$

On objects: the same as for the dual of $\mathbb{V}$ on Stone (Abramsky, Johnstone, Kupke, Kurz, Venema, Vosmaer):

$$\begin{array}{ccc} X & \mathbb{V}(X) & \\ \hline A & \mathbb{K}(A) & \end{array} \quad \text{Stone duality}$$

$$\mathbb{K}(A) = \frac{\mathrm{Free}_{\mathsf{BA}}(\{\Box_a, \Diamond_a \mid a \in A\})}{\{\Box \text{ preserves finite meets}, \Diamond = \neg\Box_\neg\}}$$

On morphisms:

$$\begin{tikzcd} X \arrow[d, tail, "R"] \\ Y \end{tikzcd} \qquad \begin{tikzcd} \mathbb{V}(X) \arrow[d, tail, "\mathbb{V}^{\mathsf{R}}(R)"] \\ \mathbb{V}(Y) \end{tikzcd}$$

$$\begin{tikzcd} B & \mathbb{K}(B) = \mathrm{Free}_{\mathsf{BA}}(\{\Box_b, \Diamond_b \mid b \in B\})/\!\sim \\ A \arrow[u, tail, "S"] & \mathbb{K}(A) = \mathrm{Free}_{\mathsf{BA}}(\{\Box_a, \Diamond_a \mid a \in A\})/\!\sim \arrow[u, tail, "\mathbb{K}^{\mathsf{S}}(S)?"'] \end{tikzcd}$$

We shall describe when an element α of $\mathbb{K}(A)$ is $\mathbb{K}^{\mathsf{S}}(S)$-related with an element β of $\mathbb{K}(B)$.

Proposition

Given a Boolean algebra A. Every $\gamma \in \mathbb{K}(A)$ is (effectively) equal to

- (DNF) a finite join of elements of the form

$$\Diamond_{a_1} \wedge \cdots \wedge \Diamond_{a_n} \wedge \Box_b$$

  with each $a_i \leq b$;

- (CNF) a finite meet of elements of the form

$$\Diamond_c \vee \Box_{d_1} \vee \cdots \vee \Box_{d_m}$$

  with each $c \leq d_j$.

$$\begin{tikzcd} A \arrow[d, tail, "S"] & \mathbb{K}(A) \arrow[d, tail, "\mathbb{K}^{\mathsf{S}}(S)?"] \\ B & \mathbb{K}(B) \end{tikzcd}$$

Enough to describe when

$$(\Diamond_{a_1} \wedge \cdots \wedge \Diamond_{a_n} \wedge \Box_b)\ \mathbb{K}^{\mathsf{S}}(S)\ (\Diamond_c \vee \Box_{d_1} \vee \cdots \vee \Box_{d_m})$$

with $a_i \leq b$ and $c \leq d_j$.

$$\begin{tikzcd} A \arrow[d, tail, "\leq"] & \mathbb{K}(A) \arrow[d, tail, "\leq"] \\ A & \mathbb{K}(A) \end{tikzcd}$$

(With $a_i \leq b$ and $c \leq d_j$:)

$$(\Diamond_{a_1} \wedge \cdots \wedge \Diamond_{a_n} \wedge \Box_b) \leq (\Diamond_c \vee \Box_{d_1} \vee \cdots \vee \Box_{d_m})$$

$$\Updownarrow$$

$$(\exists i : a_i \leq c) \text{ or } (\exists j : b \leq d_j).$$

Key idea: $\Diamond$-with-$\Diamond$ or $\Box$-with-$\Box$. [Cederquist, Coquand, 1998]

E.g.: if A, B, C, D are clopens of a Stone space X with $A \subseteq C$ and $B \subseteq D$, then

$$\Diamond A \cap \Box B \subseteq \Diamond C \cup \Box D \iff A \subseteq C \text{ or } B \subseteq D.$$

$$\begin{tikzcd} A \arrow[d, tail, "S"] & \mathbb{K}(A) \arrow[d, tail, "\mathbb{K}^S(S)"] \\ B & \mathbb{K}(B) \end{tikzcd}$$

(With $a_i \leq b$ and $c \leq d_j$:)

$$(\Diamond_{a_1} \wedge \cdots \wedge \Diamond_{a_n} \wedge \Box_b)\ \mathbb{K}^S(S)\ (\Diamond_c \vee \Box_{d_1} \vee \cdots \vee \Box_{d_m})$$

$$\Updownarrow$$

$$(\exists i : a_i\ S\ c) \text{ or } (\exists j : b\ S\ d_j).$$

Key idea: $\Diamond$-with-$\Diamond$ or $\Box$-with-$\Box$.

Theorem (A., Bezhanishvili, Carai, 2024)

The dual of the Vietoris endofunctor $\mathbb{V}^{\mathsf{R}} \colon \mathsf{Stone}^{\mathsf{R}} \to \mathsf{Stone}^{\mathsf{R}}$ *is the following endofunctor* $\mathbb{K}^{\mathsf{S}} \colon \mathsf{BA}^{\mathsf{S}} \to \mathsf{BA}^{\mathsf{S}}$*:*

- *On objects: it maps* A *to*

$$\mathbb{K}(A) := \frac{\mathrm{Free}_{\mathsf{BA}}(\{\Box_a, \Diamond_a \mid a \in A\})}{\{\Box \text{ preserves finite meets}, \Diamond = \neg\Box\neg\}}$$

- *On morphisms: it maps a subordination* $S \colon A \looparrowright B$ *to the unique subordination* $\mathbb{K}^{\mathsf{S}}(S) \colon \mathbb{K}(A) \looparrowright \mathbb{K}(B)$ *satisfying "*$\Diamond$*-with-*$\Diamond$ *or* $\Box$*-with-*$\Box$*".*

"$\Diamond$-with-$\Diamond$ or $\Box$-with-$\Box$": (With $a_i \leq b$ and $c \leq d_j$:)

$$(\Diamond_{a_1} \wedge \cdots \wedge \Diamond_{a_n} \wedge \Box_b)\ \mathbb{K}^{\mathsf{S}}(S)\ (\Diamond_c \vee \Box_{d_1} \vee \cdots \vee \Box_{d_m})$$

$$\Updownarrow$$

$$(\exists i : a_i\ S\ c) \text{ or } (\exists j : b\ S\ d_j).$$

An application

De Vries duality is a duality for compact Hausdorff spaces, which associates to a compact Hausdorff space X the Boolean algebra of regular opens, together with the binary relation $\prec$ of well-insideness: $U \prec V \iff \mathrm{cl}(U) \subseteq V$.

Question (Bezhanishvili, Bezhanishvili, Harding, 2015)

What is the De Vries dual of the Vietoris endofunctor on compact Hausdorff spaces?

We piggyback on the duality between $\mathbb{V}^{\mathsf{R}} \colon \mathsf{Stone}^{\mathsf{R}} \to \mathsf{Stone}^{\mathsf{R}}$ and $\mathbb{K}^{\mathsf{S}} \colon \mathsf{BA}^{\mathsf{S}} \to \mathsf{BA}^{\mathsf{S}}$.

Theorem (A., Bezhanishvili, Carai, 2024)

The de Vries dual of the Vietoris endofunctor on KHaus *is obtained by applying* $\mathbb{K}^{\mathsf{S}}$ *(= the dual of* $\mathbb{V}^{\mathsf{R}} \colon \mathsf{Stone}^{\mathsf{R}} \to \mathsf{Stone}^{\mathsf{R}}$*), followed by a(n appropriate) MacNeille completion.*

$$X \xrightarrow{R} Y \qquad\qquad (B, \prec_B) \xleftarrow{S} (A, \prec_A)$$

$$\begin{tikzcd}
 & (\mathbb{K}(B), \mathbb{K}^{\mathsf{S}}(\prec_B)) \arrow[d, hook] & (\mathbb{K}(A), \mathbb{K}^{\mathsf{S}}(\prec_A)) \arrow[l, "\mathbb{K}^{\mathsf{S}}(S)"'] \arrow[d, hook] \\
\mathbb{V}(X) \arrow[r, "\mathbb{V}^{\mathsf{R}}(R)"] & \mathbf{M}(\mathbb{K}(B), \mathbb{K}^{\mathsf{S}}(\prec_B)) & \mathbf{M}(\mathbb{K}(A), \mathbb{K}^{\mathsf{S}}(\prec_A)) \arrow[l, "\mathbf{M}(\mathbb{K}^{\mathsf{S}}(S))"']
\end{tikzcd}$$

where $\mathbf{M}$ is an appropriate MacNeille completion functor.

Conclusions

Key ideas

1. Beyond functions; closed relations between Stone spaces ($\leftrightarrow$ subordinations between Boolean algebras).
   Especially: in dualities between "KHaus"-like and "lattice+proximity"-like structures.
   (Scott, Vickers, Jung, Sünderhauf, Moshier, Kegelmann, Kurz, ...)
2. "$\Diamond$-with-$\Diamond$ or $\Box$-with-$\Box$" (Cederquist, Coquand):

$$\left(\bigwedge_i \Diamond_{a_i}\right) \wedge \Box_b \leq \Diamond_c \vee \left(\bigvee_j \Box_{d_j}\right) \Leftrightarrow (\exists i : a_i \leq c) \text{ or } (\exists j : b \leq d_j).$$

3. Our packaging of these ideas:
   - Stone dual description of $\mathbb{V}^{\mathsf{R}} \colon \mathsf{Stone}^{\mathsf{R}} \to \mathsf{Stone}^{\mathsf{R}}$;
   - de Vries dual description of $\mathbb{V} \colon \mathsf{KHaus} \to \mathsf{KHaus}$ and for relations.

M. Abbadini, G. Bezhanishvili, L. Carai.
Vietoris endofunctor for closed relations and its de Vries dual.
Topology Proceedings, 2024. Links: paper, arxiv.